MOLECULAR RENDERING

Molecular rendering软件

USER MANUAL

Molecular visualization and trajectory analysis

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Support: 3175911862@qq.com

github.com/wangmumu128/Molecular-rendering

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01 Overview and scope

Molecular rendering V1.6.4 is a Windows desktop application for molecular structures, simulation cells, trajectories and numerical post-processing. The workflow is import, verify types and units, configure the view, run an analysis, inspect results and export. The initial viewport is empty; no demonstration model loads automatically.

Feature coverage The application provides topology-assisted import, element recognition and colors, type selection, radius and material editing, transparent water surfaces, multiple gel groups, frame-local ice types, distance bonds, clipping, charge coloring, four viewports and three lights. Outputs include PNG/TIFF images, direct viewport captures and MP4. Analyses include RDF, MSD, density, shear stress, local shear strain, hydrogen-bond statistics and ordinary contact TCF.

Display versus scientific data Colors, display radii, materials, hiding, display bonds and reconstructed surfaces do not rewrite the input trajectory. Analyses read original particle data rather than the visually clipped subset. Select analysis types and regions explicitly. Temporary ice display types are not persistent trajectory types.

The program does not run molecular dynamics, reconstruct missing forces or atomic stresses, or infer material properties directly from a color map. Check units, identifiers, cells, boundary conditions and the meaning of each observable.

About this manual This manual describes the V1.6.4 installer, dated 2026-10-29. Screenshots use synthetic demonstration data. Example plots illustrate functions and are not research findings. The English manual covers the same features; the application interface is primarily Chinese, so Chinese control names are retained where useful.

02 Installation, activation and updates

Requirements The installer targets 64-bit Windows 10/11 and bundles its runtime; a separate Python installation is unnecessary. A working graphics driver is required for the viewport. At least 16 GB RAM and 4 GB installation space are suggested, not universal minimums. Large trajectories, fine surface meshes and large renders need additional memory and output space.

Install and activate 1. Download MolecularRendering-1.6.4-Windows-x64-Setup.exe and SHA256SUMS.txt from the official repository Releases and verify the checksum. The installer is not commercially code-signed; Windows may identify an unknown publisher.

2. Run the installer, choose a destination folder and optionally create the water-molecule desktop shortcut. The default location is the current user's local programs directory. Keep the main executable, worker and _internal directory together.

3. Send the order details and displayed device code to 3175911862@qq.com. Paste the complete activation code beginning with MR1- and activate. Existing .mrlicense files remain supported. Do not post activation codes, orders or device codes in public issues.

Personal license and updates The personal edition costs CNY 64 and covers up to two computers for the same purchaser, migration through support and all subsequently released version updates. Authorization is verified locally. Stop using the old device after migration. Save configurations and results, close the application, and run the new installer to update. Reinstalling Windows or changing device identity may require reauthorization.

Uninstall through Windows application management or the supplied uninstaller. Uninstalling does not deliberately remove the user profile's licenses, settings or results. Consult the bundled personal-use terms and third-party notices for the applicable license conditions.

03 Workspace and navigation

Workspace areas The top toolbar contains Import, Fit View, Single/Four Views, Enlarge View Area, Pick Atom, Render Preview and Export Image. The central viewport displays the model; the render-results tab displays exported images. The right panel has 修改与显示 (Modify and display) and 渲染设置 (Render settings). Select a tree item to open its parameter panel. Playback, data information and logs appear below.

Camera interaction Drag with the left mouse button to rotate, the right button to pan, and use the wheel to zoom. Enable 选取原子 or Shift-click to select an atom. 适合视口 fits the model. Drag the horizontal and vertical splitters to change the available view width and height. 放大视图区 or F11 expands the view area.

In four-view mode, the blue title identifies the active viewport. Rendering and direct capture use this view; the pointer position alone does not identify the export camera. Check the active title and output-aspect guide before exporting.

Applying edits Completed, valid display edits update the viewport automatically. An incomplete numeric entry can defer an update; press Enter or leave the editor to finish. A changed file path must be explicitly opened using import, a file chooser or Enter. Scientific results are not recalculated whenever a display parameter changes.

The figure shows the V1.6.4 workspace with a synthetic demonstration structure; this model does not load at startup.

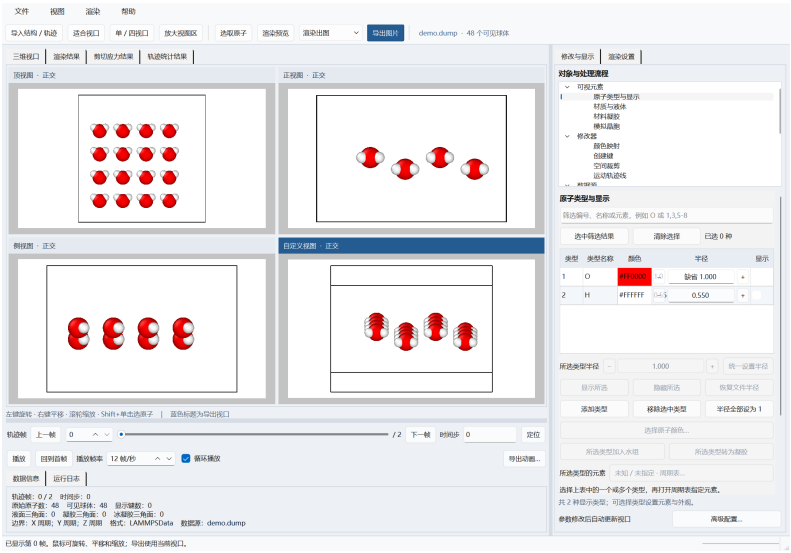


Figure 3. Workspace and navigation: synthetic demonstration or application controls.

04 Importing structures and trajectories

Input formats Supported readers include LAMMPS dump/data, XYZ/extended XYZ, CIF, PDB, XTD/XSD, POSCAR, XSF, GRO, CAR/ARC and MOL/SDF. Compatibility depends on actual file contents and the reader; renaming an extension is not a format conversion. A static structure is treated as one frame; multi-frame files support playback and range analysis.

Procedure Use 导入结构/轨迹 or drag in a local file. Check frame count, atom count, types, coordinates, cell and periodic flags. Frame indices start at zero. A simulation timestep is a separate attribute: frame 3002 does not automatically mean 3002 ps.

For LAMMPS data, the importer prefers declarations such as Atoms # full or Atoms # atomic. If the style is missing or ambiguous, choose the actual atom_style in 数据源. The full style includes molecular identifiers and charge; do not force it into atomic format. Use the generating simulation script to determine the style.

Plain XYZ can lack a cell, stable identifiers, masses or element names. Coordinates may display correctly while RDF, periodic unwrapping or mass density remain unavailable. Extended XYZ can carry a cell and additional properties, but verify the imported values. Periodic flags represent physical boundary conditions and must not be changed merely to bypass validation.

Large files The first import of a large text trajectory requires a frame index. Logs separate reading/indexing from initial processing/display time. The session caches at most two unchanged source indexes and releases them on exit. Import pauses playback; requests made during processing are deferred according to application state. Do not overwrite a trajectory while it is being read.

05 Topology assistance and element recognition

When types are absent Importing a trajectory without a particle-type column prompts for a matching topology data file. Use the same simulation and particle population. Types and identifiers can come from topology while positions follow trajectory frames. Canceling the selection preserves the current scene.

When types exist but elements are unknown Open 数据源 → 分子结构/晶胞/轨迹 **and select** 选择拓扑**. The application refreshes type names, element evidence and standard colors. Existing radii, materials, hidden states and manually assigned elements are retained. Different force-field types of the same element keep distinct type identifiers.**

When both files contain identifiers, they must be unique and their sets and counts must match; reordered rows are supported. Mismatched counts, duplicate identifiers or different identifier sets are rejected. If either file lacks identifiers, matching uses file order and logs this assumption. You must confirm that both files and every frame preserve the same order; equal counts alone do not prove correspondence.

Recognition evidence Explicit element properties and recognized names take priority. A narrow atomic-mass match provides only a candidate. Numeric type identifiers do not encode elements. Coarse-grained beads, united atoms, isotopes and reduced-unit masses must not be treated as ordinary atoms solely from mass. Mass-based assignments are marked for verification. Conflicting or insufficient evidence remains unknown; assign an element manually using the periodic table.

Topology can supply a missing fixed cell. A variable-cell trajectory must contain the true cell for each frame. Plain XYZ plus one data file cannot reconstruct missing cell history. Never replace the simulation cell with the bounding box of particle coordinates.

06 Particle types, colors, radii and visibility

Type table and multiple selection In 原子类型与显示, edit names, colors, radii and visibility. Ctrl-click selects multiple rows; Shift selects a range. Batch controls change radius, visibility, element or material. Pick an atom to identify its type, then edit the appearance of all spheres belonging to that type.

Elements and colors The periodic-table buttons assign an element to selected types. Defaults include red oxygen, white hydrogen and gray carbon, with a fixed palette for 118 elements. Unknown types receive random initial colors. A color button opens a picker with swatches, RGB/HSV controls and #RRGGBB input. 统一原子配色 restores default colors for recognized elements without changing radii or materials.

Radius modes Valid per-particle radii from the input take priority initially; missing or invalid values fall back to 1. In file mode, the displayed radius control represents the fallback, not necessarily every particle's actual radius. Editing it manually changes the type to a uniform display radius.

The plus/minus buttons and focused mouse wheel adjust radius by 0.5 per step. Apply a uniform value to multiple selected types or use 恢复文件半径 to restore per-particle values. Setting all radii to 1 changes appearance, not coordinates, cell dimensions or force-field parameters.

Hidden types remain in original-data analyses. Exclude them explicitly in RDF, MSD, stress or other analysis groups when required. Saving a display configuration retains names, elements, colors, radius modes and visibility.

07 Individual and batch materials

Individual types In 材质与液体, select a target type and preset: ordinary sphere, matte, glossy plastic, metallic, refractive glass or translucent. New spheres start matte. Roughness ranges from 0.001 to 1, opacity from 0 to 1, and the glass refractive index from 1.01 to 3. A preset supplies starting values that can be edited.

Batch procedure 1. Select the required rows in the particle-type table. Open 统一修改材质 from either the type panel or material panel.

2. Choose selected types or all types. All types includes hidden ones. Set the preset, roughness, opacity and refractive index, then confirm. Cancel leaves materials unchanged.

3. The viewport updates automatically and configuration files retain the values. Colors, radii and visibility remain independent. Frame-local ice sphere types can also be included.

Interpreting the result The interactive viewport primarily shows color and transparency. Check full reflection, roughness and refraction using render preview or an exported render. Lower roughness produces more concentrated highlights rather than simply increasing overall color brightness.

Batch material editing affects particle spheres only. Water, gel and ice-gel surfaces use their own group settings. Changing hidden sphere materials does not directly change a continuous surface that replaces those spheres.

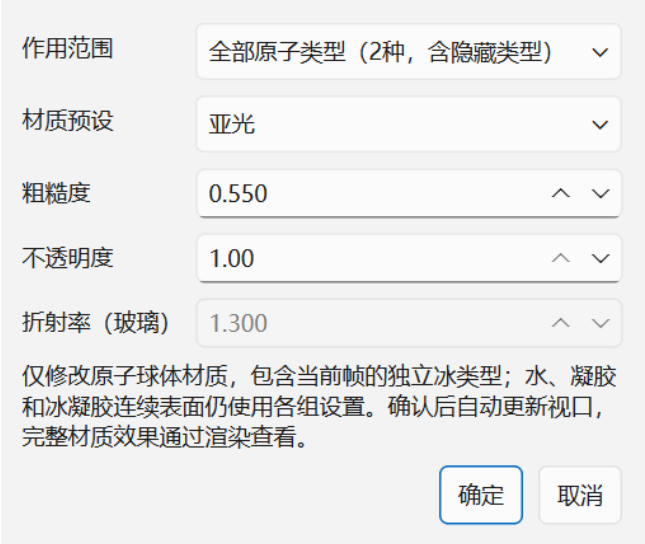


Figure 7. Individual and batch materials: synthetic demonstration or application controls.

08 Multiple water groups and transparent surfaces

Create water groups In 材质与液体, add a water group, name it and select one or more member types. A surface is reconstructed from a Gaussian density of selected positions; this is not a fluid simulation. Multiple groups have independent membership, color and geometry. Check overlapping selections to avoid representing the same particles with several surfaces.

Enable the group and choose a continuous translucent surface, surface plus ball-and-stick, or particles only. Surface-only mode hides the corresponding spheres; combined mode retains both and may appear denser.

Appearance and geometry 有色透明 keeps internal particles visible while tinting the liquid surface. 散射半透明 adds stronger diffuse scattering. Use the color picker for the surface color. A color/opacity value of zero hides the surface without deleting the group.

The smoothing radius sets the density-kernel length scale, the isovalue controls the extracted boundary, and grid resolution controls detail and computational cost. Increasing radius or decreasing threshold may connect neighboring regions, but can also merge physically separate structures. Periodic surface construction must match the actual simulation cell.

优化液体清晰度 changes appearance, lighting and sampling while retaining surface geometry, particle colors and radii. Inspect an actual render before export. Overlapping layers, coarse grids or strong scattering can reduce the visibility of internal structure.

09 Material gels and surface parameters

Workflow Open 材料凝胶, add a group and select one or more particle types. Enable the group and choose opaque gel only, gel plus particles, or particles only. Groups have independent names, activation and colors. Disabling a surface restores the corresponding particle representation.

Use the color picker for each gel. The smoothing radius is independent of sphere display radius. Isovalue, grid resolution, smoothing iterations and roughness control the extracted boundary, mesh detail, surface smoothing and reflected-light distribution respectively.

Adjustment sequence Start with a lower grid resolution to inspect membership, adjust radius and threshold to connect the intended region, and then increase detail. Larger grids and more atoms increase memory and processing time. Excessive smoothing can conceal pores and local morphology; describe the reconstruction method when using the image scientifically.

Gel surfaces are always opaque. Transparent water groups may coexist, but assign membership separately. Multiple gel groups represent selected materials or type combinations; they do not automatically identify distinct chemical phases.

This display operation does not simulate gelation or calculate physical surface tension, porosity or material volume fractions. Save membership, smoothing radius, threshold and resolution with research figures for reproducibility.

10 Frame-local ice identification and ice gel

Identification steps Create a water group, select the target frame and open 当前帧冰相识别·CHILL+. Select water oxygen types, or one coarse-grained center per water molecule. Exclude hydrogens and substrate atoms. Centers must belong to the selected water group and number at least five. The default cutoff is 3.5 in input coordinate units, a starting value for common atomistic water in angstroms, not a universal setting.

Identification creates independent display groups/types for hexagonal ice, cubic ice and optionally interfacial ice. Other structures are not automatically liquid water. Hydrates and interfacial hydrates are excluded from ice groups. Reliable positive molecular identifiers allow associated water-group atoms to follow their center. Multiple centers sharing an identifier are classified individually, with a warning rather than ambiguous membership expansion.

Appearance Edit ice type names, elements, colors, radii, materials and visibility independently in the particle table. Source types remain distinct; oxygen and hydrogen are not merged. Remaining particles retain their original water types and transparent surfaces.

Selected ice groups may display as opaque continuous ice gel with separate color, radius, threshold, resolution, smoothing and roughness settings, or as spheres. Hiding ice does not return its members to the liquid surface.

Validity Identification belongs only to the current frame and raw data. Changing frames clears temporary ice types and restores water types; returning also requires re-identification. Loaded configurations check frame/data/group consistency. Animation export does not automatically identify ice in every frame.

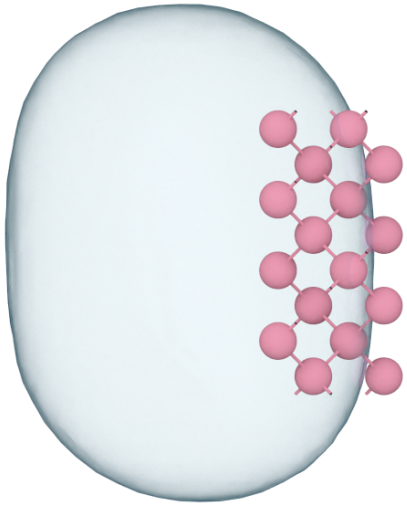


Figure 10. Frame-local ice identification and ice gel: synthetic demonstration or application controls.

11 Bonds, clipping and color mapping

Display bonds 创建键 offers no bonds, distance rules by type pair, or input bond topology. Automatic initialization uses 1.2 times the sum of the two recognized elements' covalent radii; O-H starts at 1.164 angstrom. Choose angstrom or nm before generating. Other or reduced units require manual distances.

Double-click a cutoff to edit it. Ctrl/Shift-select rows for batch deletion or clear all generated rules. Changing units does not convert existing manual values. Unknown elements, noble gases and elements without radius records are skipped. Distance bonds do not infer bond order, reactions or force-field connectivity, rewrite input Bonds, or supply topology for hydrogen-bond analysis.

Clipping and simulation cell Enable 空间裁剪 and enter Cartesian X/Y/Z bounds. Particle centers outside the range and their associated bonds are hidden. Valid bounds are required. This display filter does not change analysis input. The cell/camera panel shows or hides the box and adjusts viewing direction; it does not redefine the physical input cell.

Color mapping Type coloring uses recognized or custom colors. Charge coloring needs charge values from the file and a valid color range; colors cannot reconstruct absent charges. Current-frame shear-strain coloring requires a completed strain calculation for that frame and cannot reuse values from another frame.

Save element evidence, distance units, cutoffs and color ranges with the display configuration. Use consistent settings when comparing frames so a display change is not mistaken for a structural change.

12 Camera, lights and shadow depth

Camera and composition Use mouse rotation, pan and zoom, or set azimuth, elevation and relative fit zoom numerically. Select the blue-titled view in four-view mode. Output width and height determine the render aspect ratio, which may differ from the screen window. Inspect the aspect guide and actual render preview.

Styles and manual control 明亮显色 emphasizes color visibility, 厚重阴影 increases contact depth, and 手动光照 retains custom lighting. Shadow strength and contact shadows adjust local depth. Reducing every light merely darkens the whole image; it is not equivalent to strengthening contact contrast.

Enable three-light direction control and open 球形预览与光源调节. Select the light to move and adjust its direction in the sphere preview. Enable and set intensity independently for each light. Ambient illumination fills shaded regions separately from directional lights. Start with front key, fill and lower-front fill lights, then inspect the actual model.

Important behavior 应用正面光照与亚光 replaces current lighting and all sphere materials while retaining colors, radii, names, camera and liquid settings. It does more than change camera direction. Offline preview/export uses rendering lights; the interactive viewport uses real-time display illumination, so their appearance can differ.

Fix the camera and background first, tune lights individually, then adjust ambient fill and contact shadows. Check bottom-layer highlights, transparent surfaces and internal colors in actual rendered output rather than relying only on the small sphere preview.

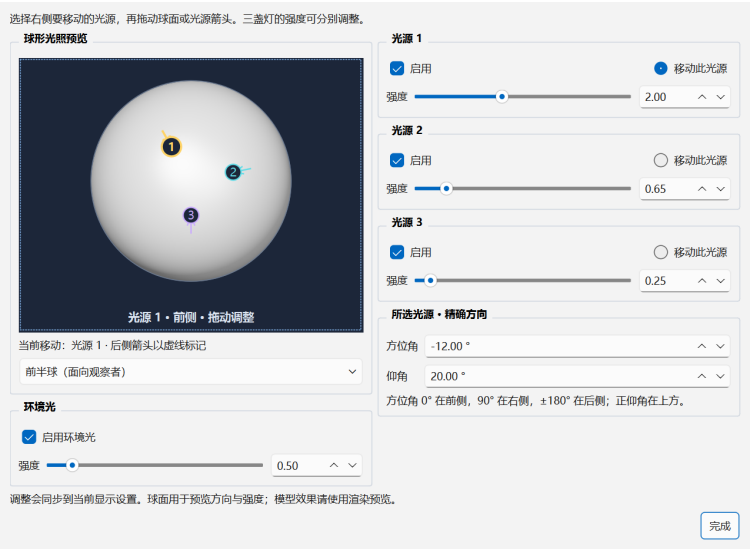


Figure 12. Camera, lights and shadow depth: synthetic demonstration or application controls.

13 Still rendering, DPI and direct capture

Offline rendering Choose 渲染出图 in Render settings, then set width, height, DPI, format and filename. Defaults are 3840 × 2160 pixels. Each dimension accepts 128-16000 pixels; DPI accepts 600-2400. PNG and TIFF are lossless and support transparent backgrounds. White, black or custom background colors are available.

Run a small ray-traced preview to inspect framing, materials, lights, transparent surfaces and legends before exporting. More samples generally increase computation. Denoising is a separate post-render step. The application does not retain a separate undenoised image copy; consult logs/output records to determine whether denoising actually completed rather than relying on the progress bar alone.

What DPI means DPI sets print size, not pixel detail. Printed width in inches equals pixel width divided by DPI. A 3840-pixel image is 6.4 inches wide at 600 DPI and 1.6 inches at 2400 DPI. Increase pixel dimensions or sampling when more detail is needed.

Direct viewport output Choose 视图直接出图 or use the Render menu / Ctrl+Shift+R. The application pauses playback and captures the active viewport framebuffer without offline rendering or denoising. Pixel dimensions depend on viewport size and operating-system scaling; enlarge the window/view area for a larger capture.

Direct capture retains the visible background and internal margins but excludes toolbars and settings panels. It does not add transparent background or an offline-only color legend. Render width/height and transparency settings do not apply. Blank filenames are generated automatically; repeated names receive a suffix. Do not enter paths or prohibited filename characters.

14 Playback, animation and particle trails

Playback Use previous/next frame, the frame number, slider or timestep locator. Playback rate, looping and return-to-first-frame controls are available. Frames per second sets viewing speed, not simulation timestep. Positions come from saved frames; intermediate dynamics are not simulated.

MP4 export In 轨迹动画, set start/end frames, stride, frame rate, pixel dimensions and filename, then export MP4. Width and height must be even. The active viewport camera stays fixed while current display, material, lighting and device settings are rendered frame by frame. MP4 has no alpha channel and uses the selected background color.

Cost increases with output frame count, resolution, samples and surface complexity. Test a short interval before exporting a complete sequence. Frame-local ice and shear-strain snapshots are not recalculated automatically for other frames. Do not treat a single-frame analysis as the result for an entire animation.

Particle trails In 运动轨迹线, enter stable particle IDs and choose the frame interval, stride, line diameter and color. Periodic-jump unwrapping is optional. Lines connect saved positions over the selected interval; they may extend beyond the cell and are not automatically clipped with the particle display.

Trails need stable identifiers. Long intervals or many selected IDs increase reading and display costs, so begin with a few representative particles. Missing intermediate trajectories and multiple unresolved periodic crossings cannot be reliably reconstructed from sparse positions alone.

15 Shared analysis settings

Where to start The 数据分析 tree contains shear stress, shear strain, RDF/MSD/density, hydrogen-bond TCF, hydrogen-bond density and dynamic counts. Ordinary atom-pair contact TCF is selected in the trajectory-statistics method list and is separate from hydrogen-bond TCF.

RDF, MSD, density, hydrogen-bond and ordinary TCF tasks use start/end frames and stride. Indices start at zero and a valid end frame is included. Up to 12 groups are supported; each can contain multiple types. RDF/contact TCF use two selections; hydrogen bonds use D/H/A. Type fields accept lists and ranges, for example 1,2,5-8.

Units and time Specify the actual coordinate unit; the application does not infer it from numerical magnitude. The physical interval is the time between adjacent stored frames, subsequently multiplied by the selected stride. Stored frames 0.5 ps apart with stride 2 produce samples 1 ps apart. The integration timestep must be combined with output frequency to obtain the stored-frame interval.

Execution and output Use 计算并导出600 DPI图片与TXT数据. Analysis runs in a separate process, pauses playback and locks related inputs; cancellation is available. Successful results include PNG, TXT, parameter JSON and result JSON. Failed/canceled work is not a complete result; do not interpret leftover temporary files as final scientific output.

A result is a snapshot of sources and parameters. Changing frames, groups or files does not recalculate the previous result. Visibility, surfaces and temporary ice display types do not redefine the original analysis population. Source-file changes during computation invalidate the result.

16 Radial distribution function (RDF)

Setup Choose RDF and add groups. Select center types A and neighbor types B independently; an empty neighbor selection uses the group types. Set legend names, frame interval, cutoff and bins. The bin range is 4-4000; defaults are cutoff 5 and 200 bins, with distances in input coordinate units.

Each frame counts directed center-neighbor pairs, excludes self-pairs and duplicate periodic images, and normalizes by cell volume, eligible pair count and shell volume. Overlapping A/B selections receive the self-pair correction. Frames are normalized separately and then averaged with equal weight.

Cell and boundary requirements V1.6.4 supports fully periodic 3D cells and slabs periodic in two directions with one nonperiodic direction. Slabs use each center's shell volume accessible between the two cell faces. Two or more nonperiodic directions are unsupported. The cutoff cannot exceed half the smallest periodic-direction face spacing in any selected frame. Selected particles must lie inside the nonperiodic cell bounds.

This correction concerns cell boundaries, not physical material surfaces or vacuum removal. Inhomogeneous/vacuum-containing systems still use neighbor density averaged over the entire cell. A peak value is not a coordination number. For XYZ without cell information, supply a matching fixed-cell data file or a trajectory containing each frame's cell.

Output Groups share one 600 DPI plot. TXT contains r followed by each g(r); JSON records frames, groups and boundary treatment. Test a few frames and verify types/distances before increasing the range. The example plot uses synthetic data.

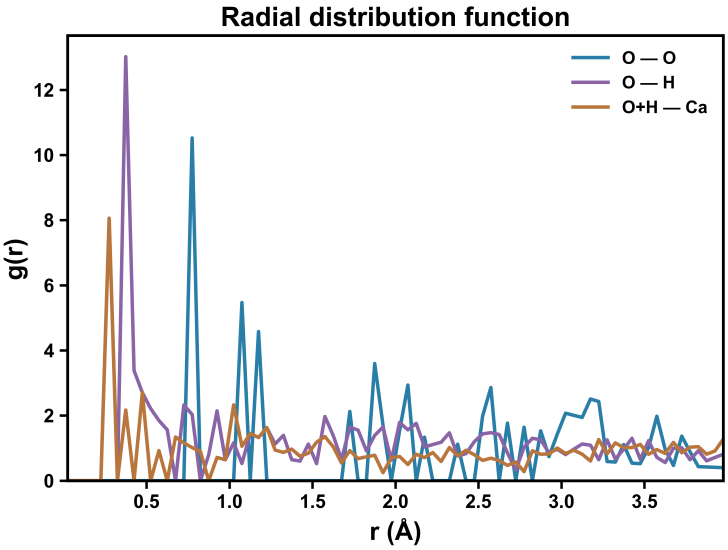


Figure 16. Radial distribution function (RDF): synthetic demonstration or application controls.

17 Mean-squared displacement (MSD)

Origins and dimensions Choose MSD and one or more particle groups. 起始帧参考 measures displacement from the selected starting frame; 多个时间起点平均 averages over available origins and particles at each lag. Choose xyz, xy, xz, yz, x, y or z. Removing each group's overall translation is optional and disabled by default.

MSD is the particle average of summed squared displacements in the selected directions, with units of length squared. It is not speed, displacement magnitude or a diffusion coefficient. Automatic diffusion fitting is not provided; any external fit must account for dimension, time units and the appropriate regime.

Identity and periodic unwrapping The selected reference-frame particles are tracked using stable, unique IDs; row order may change. Without IDs, enable fixed-order matching only after confirming unchanged particle order across frames.

Automatic processing prefers unwrapped coordinates such as xu/yu/zu, then image flags ix/iy/iz, otherwise incremental unwrapping. Incremental recovery reads all stored frames in the interval even when the output stride is larger. Sparse output can lose multiple boundary crossings irreversibly. Variable-cell trajectories should provide continuous coordinates or image flags.

The multiple-origin coordinate buffer is limited to 512 MiB. Reduce particles or sampled output frames, or use the reference-frame mode when necessary. PNG shows the selected dimension; TXT also provides directional components and origin counts. Preserve the time interval and drift-removal setting for reproducibility.

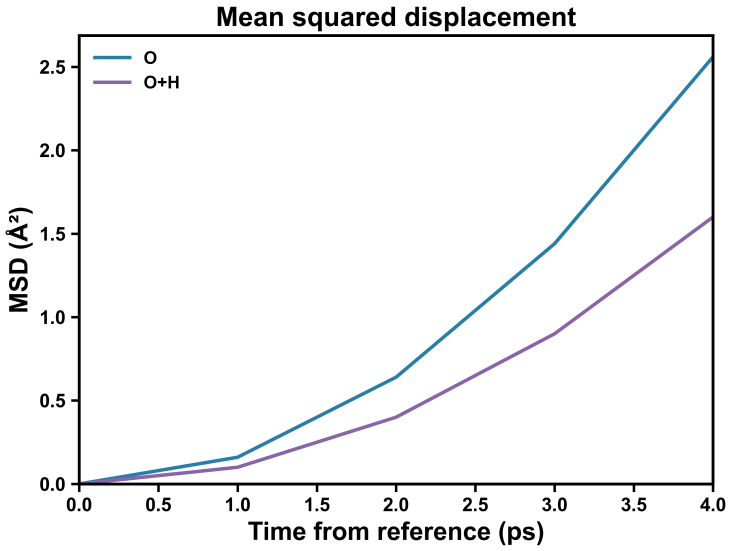


Figure 17. Mean-squared displacement (MSD): synthetic demonstration or application controls.

18 One- and two-dimensional density

Directions and coordinates Choose density, select 1D or 2D, define multi-type groups and set frames. One-dimensional directions are x/y/z. Planes include xy, xz, yz and reversed ordering yx, zx, zy; reversal changes plot-axis order, not the structure.

Cartesian coordinates require a fixed axis-aligned orthorhombic cell. Use cell fractional coordinates for tilted or variable cells: x/y/z then mean a/b/c, normally over 0-1. Normalization still uses each frame's actual volume. Set bin ranges and count; default ranges come from the cell.

Slices and observables A 2D map can restrict the normal-direction slice. An xz plane uses y thickness; fractional ac uses a b interval. The lower limit is included and upper excluded. The result is mean volume density through the chosen thickness, not areal density.

Number density is particle count divided by bin volume. Mass density requires per-particle masses, type masses or explicit overrides such as 1:15.999,2:1.008. Confirm input mass and length units for conversion to g/cm³. Display sphere size does not determine particle mass.

Outputs and comparison Normalize each frame before equal-weight averaging. 1D groups share axes; 2D groups are panels in a single PNG with a common color range. The 2D TXT is a long table of two coordinates and group densities. Compare models using consistent coordinates, slice thickness, binning and physical units.

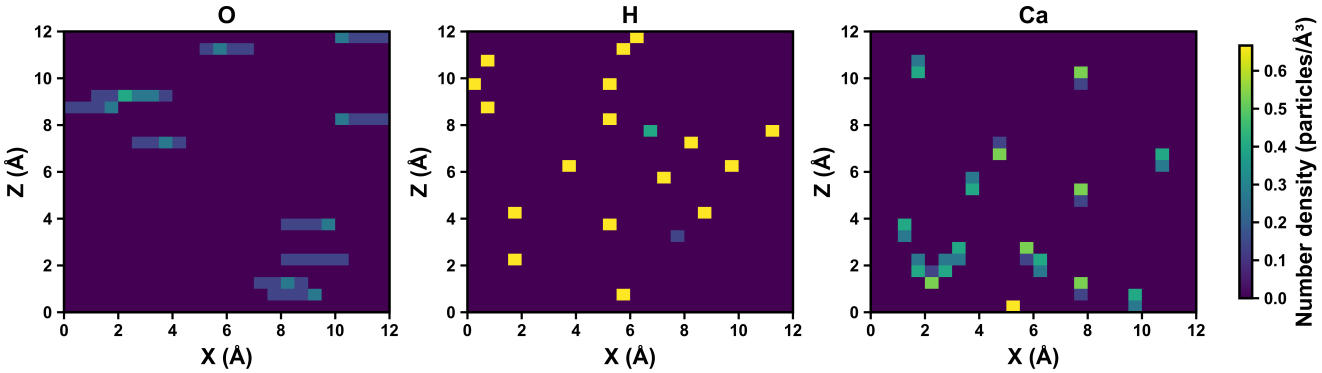


Figure 18. One- and two-dimensional density: synthetic demonstration or application controls.

19 Two shear-stress calculation routes

Preparation Open 剪切应力 and read available data fields/types. Choose atomic-stress sum divided by volume or tangential-force sum divided by area. Select all/subset types and optionally a Cartesian box, including lower and excluding upper bounds in original coordinates. Confirm units, fields, denominator and sign, then calculate a frame or interval.

Atomic-stress route tau_ab = sign × sum(S_i,ab) / V. S must have stress-times-volume units. LAMMPS stress/atom components are xx, yy, zz, xy, xz, yz, so shear terms correspond to components 4/5/6. An arbitrary fourth data column is not automatically shear stress. The default +1 retains atomic-stress sign; -1 gives the corresponding pressure-tensor sign.

V may be the instantaneous cell, a selection-box volume or a manual value. A subset normalized by the whole cell gives its contribution to whole-cell stress, not its own material-average stress. Atomic volumes are not estimated. Kinetic terms and included interactions are determined by the simulation output.

Force route tau = sign × sum(F_t) / A. Choose different tangent and normal axes and the force field. Automatic area uses the appropriate cell-face vector cross product. A tilted face not perpendicular to the chosen Cartesian normal requires a manual physical area. The application does not automatically double or halve area for multiple interfaces.

fx/fy/fz are usually net forces and may include constraints or external forces. They represent interface stress only when the selected force has that meaning; zeroed forces cannot be recovered. Supported units are real, metal, si, cgs, micro, nano and lj. Physical units output MPa; lj remains reduced. Values are instantaneous, without automatic averaging or absolute values. CSV retains sums, denominators, units, frames and timesteps.

20 Per-particle shear-strain maps

Procedure Open 剪切应变云图 and choose the current and undeformed reference frames; reference defaults to 0. Set neighbor cutoff and participating types, or leave types empty for all. Selection limits both centers and fitting neighbors. Run the current-frame calculation to activate strain coloring.

A least-squares mapping of reference neighbors to current neighbors gives local deformation gradient F. The Green-Lagrange tensor is E=(F^TF-I)/2. The plotted invariant is sqrt[Exy²+Exz²+Eyz²+((Exx-Eyy)²+(Exx-Ezz)²+(Eyy-Ezz)²)/6].

This is a nonnegative dimensionless invariant, not signed engineering shear strain or stress in MPa, and it is not automatically expressed as percent. A sufficient nondegenerate 3D neighborhood is required. Invalid fits appear gray and leave CSV values blank rather than zero. If all fits fail, the calculation stops.

Periodicity and display IDs must be unique with matching selected sets in both frames; otherwise explicitly confirm fixed order. Minimum-image treatment applies to suitable wrapped coordinates and cannot recover arbitrary displacements beyond half a cell. Uniform cell deformation is retained by default; optionally removing it changes the observable.

The default display range is 0-2.5 and can be edited or fitted to valid values. Saturated colors do not clip CSV numbers. Strain mode temporarily uses opaque matte spheres and hides continuous surfaces; restore type coloring to recover normal appearance. Results are frame-local and require recalculation after a frame change. CSV contains identifiers, coordinates, validity and strain. Keep cutoff and color range consistent across comparisons.

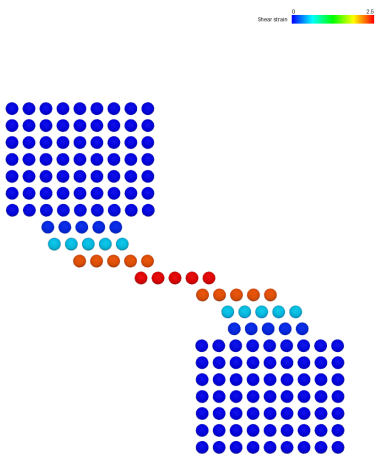


Figure 20. Per-particle shear-strain maps: synthetic demonstration or application controls.

21 Hydrogen-bond definitions and geometry

Select D, H and A Each hydrogen-bond group independently selects donor heavy atoms D, explicit hydrogens H and acceptor heavy atoms A. D and A may overlap; H must not overlap heavy-atom types. A hydrogen-free coarse-grained model cannot directly use this explicit-hydrogen criterion.

Defaults are D-A distance at most 3.5 angstrom, D-H-A angle at least 150 degrees, and D-H geometric pairing cutoff 1.2 angstrom. Hydrogen-bond distance controls use angstrom and convert using the declared coordinate unit, unlike ordinary contact TCF cutoffs in native coordinate units. The angle has its vertex at H.

Assign donors The default prefers input D-H topology and otherwise finds the nearest donor within cutoff. Topology-only and geometry-only modes are available. Equidistant candidates are not arbitrarily resolved. Automatically generated display bonds are not analysis topology. Intermolecular-only counting requires reliable positive molecular identifiers.

Each unique D-H-A triplet is one bond; different H atoms connecting the same D-A pair count separately. Overlapping groups are independent and may double-count when summed. D-H and D-A vectors use minimum-image connections in periodic directions, with no wrapping along open directions. Periodic cutoffs must not exceed half the smallest face spacing.

Dynamic counts, hydrogen-bond TCF and density share these selections and criteria. Keep D/H/A, donor pairing, distances and angle consistent across comparisons. Do not mix these results with definitions that merge bonds solely by D-A pairs.

22 Dynamic counts and hydrogen-bond density

Time-dependent counts Choose 动态氢键数量 and set groups, geometry, frames and the interval between stored frames. Each frame counts unique D-H-A triplets and plots the total against time. It is not coordination per molecule and is not divided by donor or hydrogen population. Any normalization must specify its denominator separately.

Spatial distribution Select 1D or 2D hydrogen-bond density, direction/plane, bins and optional slice. A bond is located at the midpoint of the shortest D-A periodic connection. Each frame's bond count per bin is divided by actual bin volume before averaging. Frames without bonds contribute zeros.

Directions are x/y/z; planes include xy/xz/yz/yx/zx/zy with optional normal limits. Cartesian mode requires a fixed orthorhombic cell; use fractional coordinates for tilted/variable cells. Two-dimensional output is volume density through the selected thickness, not areal density. Multi-group panels share a color scale.

Outputs and interpretation Outputs are 600 DPI PNG, UTF-8 TXT and JSON records. Group names become legends or panel titles; shorten, wrap or resize long labels in the chart editor. Groups can share atoms and triplets, so summing curves without deduplication may not yield a system total.

Local changes can reflect population, criteria, spatial redistribution or structural changes. The program reports the defined statistics; it does not infer causality or convert counts to hydrogen-bond energy.

23 Hydrogen-bond and ordinary pair TCF

Two distinct analyses Hydrogen-bond TCF follows the same D-H-A triplet. Ordinary contact TCF selects multi-type groups A/B and a cutoff; an unordered particle pair is in contact when its shortest separation is at most that cutoff. Self-pairs and duplicate images are excluded. Ordinary TCF requires no hydrogen, angle or D-H topology and is not an autocorrelation of distance values or position vectors.

Let P(t0) be the starting set. Intermittent C(k;t0)=|P(t0) intersect P(t0+k)|/|P(t0)| allows breaking and reforming. Continuous S(k;t0)=|P(t0) intersect P(t0+1) intersect ... intersect P(t0+k)|/|P(t0)| requires persistence in every sampled intermediate frame.

Time averaging Average equally over nonempty origins having enough subsequent frames. Maximum lag and origin stride use sampled-frame units. Physical time also includes the stored-frame interval and sampling stride. With no valid origins, JSON records null and TXT nan; the graph leaves a gap rather than forcing zero.

For contact, broken contact, restored contact over three frames, the two-frame-lag intermittent value is 1 and continuous value 0. Continuity refers only to sampled frames and cannot resolve unrecorded short breaks. Report the sampling stride because it can change continuous TCF.

Identifiers and outputs Stable unique identifiers are required; fixed-order correspondence is permitted only when verified. Changes in selected IDs or types stop the calculation. TXT includes valid-origin counts. Automatic lifetime fitting or extrapolated integration is not provided. If memory protection triggers, reduce frames/groups or increase stride, then reassess temporal resolution.

24 Chart text, typography and replotting

Before calculation Open 设置图表文字、字体与字号 to edit the output name, title, axis labels and 2D colorbar label. Set font family, size from 6 to 72 pt, bold and italic independently for main/panel titles, axis labels/ticks, legends and colorbar labels/ticks. Fonts come from the local system; missing characters use fallback fonts.

Legends and 2D panel titles originate from group names and can be edited. Figure text supports Ca^{2+}, H_{2}O and superscript/subscript buttons. Markup is parsed as text and does not execute arbitrary TeX. Filenames remain plain text; Unicode characters such as H_2O are permitted, but font formatting is not.

After calculation In 轨迹统计结果, choose 编辑当前图表 to change text and fonts, including individual legends or panel titles. Saving and replotting creates a new result directory, preserves the original image and does not reread or recalculate the trajectory. TXT retains original numerical data/group names, while display-name overrides are recorded in JSON.

Layout Analysis plots use 600 DPI, white backgrounds, black axes and a common multi-group palette. Final pixel dimensions adjust to labels, fonts and panel count rather than remaining fixed at 3840 × 2880. Long titles/colorbars wrap and are included in the saved bounds. Shorten labels or reduce font size rather than omitting necessary units.

Inspect exported PNGs at readable zoom for subscripts, units, legends, ticks and colorbars. Keep fonts, units, ranges and color scales consistent across publication panels, and retain TXT plus parameters alongside the figures.

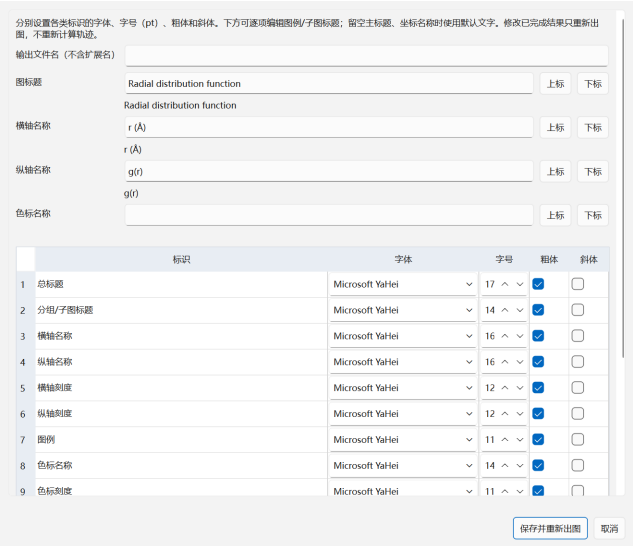


Figure 24. Chart text, typography and replotting: synthetic demonstration or application controls.

25 Configurations, result files and data management

Display configuration The source panel can load/save display configurations containing types, colors, radius modes, materials, groups, lights, output settings and analysis forms. Advanced settings edit JSON and require valid fields/values. A configuration is not a trajectory copy: when moving computers, supply original models and check type identifiers.

Locations and file roles The Windows user-data root is %LOCALAPPDATA%\MR\MolecularRendering\1. Common folders are 渲染结果 (renders), 视图出图 (captures), 轨迹分析结果 (trajectory analyses) and 剪切应力结果 (shear results). Use the File menu/result-page folder button; tasks normally use separate timestamped directories.

PNG/TIF stores still images; MP4 stores video. TXT contains UTF-8 tab-separated statistics with # comment lines. CSV stores shear stress or per-particle strain tables. 计算参数.json records sources/settings; 结果.json contains complete results/status. A parameter file or temporary .partial file alone does not indicate success.

Back up figures, data and parameters together. Replotting creates a new directory and keeps original files. Source paths, sizes and modification times support consistency checks. After moving or regenerating a source, recheck provenance rather than assuming previous results update automatically.

Privacy and licenses Trajectories and analyses are processed locally; automatic model upload is not configured. Users voluntarily send device codes to support. Review paths/research content before sharing logs, and never publish authorization files or personal orders. Application terms and third-party notices are installed with the program. Public-document licensing is not a license to the application's proprietary code.

26 GPU, CPU and performance

Rendering devices Offline rendering offers Auto, CPU only and GPU only. Auto tries a compatible NVIDIA CUDA/OptiX backend and attempts CPU if unavailable or rendering fails. GPU-only reports failure instead of silently changing device. Logs and output JSON record actual device and fallback reason.

An AMD/Intel GPU or an incompatible CUDA setup does not imply support for this offline backend. The interactive viewport uses the graphics driver through a different path. For small jobs, initialization/compilation can outweigh GPU savings.

Analysis devices Trajectory statistics offer GPU priority or CPU only. Compatible CUDA kernels accelerate selected distance, angle, histogram and reduction operations. Reading, neighbor search, identifier matching, unwrapping and some FFT/sparse-set work remain on CPU. Automatic fallback is recorded; GPU priority does not mean all analysis stages run on GPU.

Control workload Validate with a few frames, lower surface resolution and a small render preview first. Increase frame count, pixel dimensions and samples for final output. Fine water/gel meshes, overlapping surfaces, high sampling and large images increase memory/time. Smaller displayed spheres do not necessarily reduce neighbor-search cost.

Large text files need an initial frame-index scan; unchanged sources can reuse the session cache. Multiple-origin MSD and TCF enforce memory limits. Respond by reducing the scientifically appropriate workload rather than blindly raising limits. Retain actual device, sampling and mesh settings when comparing performance.

27 Troubleshooting

RDF reports a missing 3D cell Plain XYZ coordinates do not determine physical volume or periodic distances. Supply matching data for a fixed-cell trajectory, or per-frame cell metadata for variable-cell data. Do not use coordinate bounds as a cell. V1.6.4 identifies the actual failed analysis instead of labeling every error as shear stress.

Topology mismatch or wrong element colors Verify simulation identity, particle counts and identifier sets; check order if IDs are absent. Different force-field types can represent the same element. Verify mass candidates; coarse-grained types may need manual assignment or remain unknown. Explicit manual elements take precedence. Standard recoloring applies only to assigned elements.

LLVM initialization, bad allocation or rendering failure Keep the complete log. Test CPU-only, a smaller image, fewer samples and coarser surfaces. Check RAM/VRAM, graphics drivers and writable output space. If CPU also fails, provide a minimal reproducible model/settings rather than suppressing errors or changing the physical cell. Inspect rendering and denoising stages separately when denoising fails.

Dark/overexposed images or clipped labels Confirm the active view and intended style, then inspect lights, ambient fill, contact shadows, materials and overlapping surfaces. Evaluate actual rendered output. Edit chart fonts, sizes and wrapping and replot when labels are incomplete. Increasing DPI does not recover missing pixels.

Empty or unexpected statistics Check group population, units, periodic cutoff limits, stable identifiers and reference frames. No valid TCF origins produces nan; invalid strain fits are gray, not zero. For support, send version, steps, non-sensitive parameters and logs to 3175911862@qq.com.

28 End-to-end workflows and checks

Workflow A: color a trajectory and export Import the trajectory. Supply matching data when types are missing, or attach topology in the source panel for numeric types lacking element evidence. Verify counts, IDs and element candidates. Batch-edit radii/materials, create display bonds, set the active camera/lights, preview, then export PNG/TIF at 3840 × 2160 and 600 DPI. Save configuration and output records.

Workflow B: multi-group RDF Define named center/neighbor pairs. Verify cells and periodic flags and test a short interval with appropriate cutoff/binning. Inspect curves, TXT columns, units and normalization before expanding the range. Unify legend/axis typography using the editor and retain original data. A short validation run does not replace the full requested interval.

Workflow C: water, ice and gel Create a water group and inspect transparency. On the target frame, run CHILL+ using water oxygens or unique coarse-grained centers. Edit independent ice sphere colors/materials or switch to opaque ice gel, leaving remaining water transparent. Re-identify after changing frames and record cutoff, groups, frame and surface settings.

Workflow D: contact persistence For ordinary TCF select A/B and contact distance; for hydrogen-bond TCF select D/H/A and geometry. Verify IDs, stored-frame time interval, stride, maximum lag and origin stride. Export continuous/intermittent functions together and inspect origin counts/nan values. Specify any separately applied lifetime-fitting method.

Delivery checks Confirm that plots and TXT/CSV refer to the same frames/settings; filenames, units, legends and colorbars are complete; DPI and pixels meet the purpose; illustrative surfaces are distinguished from quantitative results; and sources, devices and criteria are traceable. Repository: https://github.com/wangmumu128/Molecular-rendering. Support: 3175911862@qq.com.